Department of Computer Science

CS5702 Modern Data

Postgraduate Study Guide for 2026/27

TABLE OF CONTENTS

Module Details 1
Access to Support Material and Additional Information 1
Introduction/Aims/Background 1
Learning Outcomes 2
Method of Teaching 2
Lecture/Seminar/Lab Programme 3
Reading List 3
Assessment 4
Expectations of Artificial intelligence Use 4
Additional Vital Information 5

MODULE DETAILS

Module Leader	Prof Martin Shepperd / Dr Ferdoos Hossein Nezhad	
Department	Computer Science	
Credits	n.a. (study block only)	
Other staff	Dr Valeriia Haberland	
Contact and private study time	Lectures	9 hours
	Labs/Seminars	18 hours
	Coursework Study	50 hours
	General Study	73 hours
	Total	150 hours
Assessment	Jointly with CS5701, this is covered by the assessment blocks **CS5801** Quantitative Data Analysis (coursework) and **CS5802** Critical Analysis of Modern Data (2 hour closed book).	

ACCESS TO SUPPORT MATERIAL AND ADDITIONAL INFORMATION

Most teaching, learning and support material is provided electronically via the University's Brightspace system. Note that the details provided in this study guide are based on the formal module syllabus for this module which sets out the agreed content, learning outcomes, assessment and teaching methods. Module syllabus and scheme of studies documents for your programme of study can be found by on the University's Quality Assurance web pages.

INTRODUCTION/AIMS/BACKGROUND

The aim of this module is to:

> *Provide an introduction to data management and exploration. This entails an appreciation of the richness and availability of different data sources coupled with an overview of techniques, methods and processes for **modern data** analysis. The students will learn to plan and undertake a data analytics project. The concepts of data quality, management and retrieval will be discussed. Strategies and approaches to data preparation and cleaning (often referred to as wrangling) will be presented. The importance of adopting a disciplined and engineering approach (rather than hacking) is emphasised. Exploratory data analysis (EDA) methods and data visualisation techniques will be introduced.*

Brunel University of London

Throughout this study block our approach will be hands-on and practical in nature. We will use the R-ecosystem as our primary toolset. R is a widely used programming language in data science.[1]

Note that this module comprises this study block (CS5702) and separate, but associated, assessment blocks (CS5801 and CSCS802) which are shared with CS5701 (Quantitative Data Analysis). The rationale for shared assessment blocks is that it makes little sense to think about data without considering what it will be used for, and it makes little sense to do statistical analysis without considering the underlying data. This documentation refers to the study block 'Modern Data'.

Learning Outcomes

Whatever module or programme of study you are studying for at Brunel University of London, there are learning outcomes (LO) that you must meet/achieve in order to be awarded the credits which comprise the module and programme of study. In order to get a pass grade (C- or above) in this module, you must meet the learning outcomes below, that is, you must demonstrate ability to:

LO1: Develop understanding of modern data and identify relevant methods and technologies for data management, data retrieval and data sharing.

LO2: Critically evaluate the quality of data and plan the steps required for data preparation and management.

Method of Teaching

Our approach to teaching will be predominantly practical with an emphasis upon using tools and techniques to solve real-world data problems. In general, you will find all the material you need, or links to the material, on Brightspace, the university virtual learning environment (VLE)and the online module book Modern Data.

[1] R Core Team. 2020. "R: A Language and Environment for Statistical Computing." https://www.R-project.org/; R Foundation for Statistical Computing, Vienna, Austria.

Brunel University of London

LECTURE/ LAB PROGRAMME

This module has been scheduled to run during Semester 1 and the lectures and labs are scheduled as follows:

Semester 1

Date	Week	Lecture Topic	Lecturer	Lab Activity
23/09/2026	0	Induction material for 'newbies'		
30/09/2026	1	Module overview, motivating examples, the R ecosystem	MS	What is data science? What can we do with R? Basic R and descriptive stats.
07/10/2026	2	The richness of data: structured and unstructured	MS	Analysing the module survey.
14/10/2026	3	Engineering or hacking? Readable code and reproducible data analysis	MS	Finding and importing data. Simple visualisation of time series, smoothing.
21/10/2026	4	Exploratory data analysis (EDA) and visualisation	FHN	Exploring data: more complex questions (salary and gender analysis)
28/10/2026	5	Data quality, cleaning and imputation	VH	Cleaning data set examples. Imputation with R packages.
04/11/2026	6	READING WEEK		
11/11/2026	7	Presenting data effectively	MS	Generative AI and your workflow.
18/11/2026	8	NLP and text processing	FHN	Text mining, word clouds.
25/11/2026	9	Sentiment analysis	FHN	Sentiment analysis in R.
02/12/2026	10	Data dimensionality reduction	FHN	Exploring High-Dimensional Data with PCA and Autoencoders in R
09/12/2026	11	Exam (CS5802) revision	MS	Course work support.

Please note: Week 6 in Semester 1 and Week 23 and 27-29 in Semester 2 are weeks during which there are no scheduled lectures, labs, seminars or tutorials. Weeks 15-17 are the Winter Examination Period. Week 18 is the first week of teaching after the Christmas break.

READING LIST

Core reading list

Kabacoff, R., 2022. R in Action: Data analysis and graphics with R and Tidyverse, 3rd Edn., O'Reilly Safari: link. (There are also associated videos here.)

Peng, R.D. and Matsui, E., 2015. The art of data science. A Guide for Anyone Who Works with Data. Skybrude Consulting, LLC, download from https://bookdown.org/rdpeng/artofdatascience/

Zumel, N., Mount, J. and Porzak, J., 2019. *Practical Data Science with R*. 2nd Edn., O'Reilly Safari: http://tinyurl.com/yxjhwvuf (There are also associated videos but based on the 1st Edn here.)

Supplementary Reading

Online / free books

Grolemund, G. 2014. Hands-On Programming with R online at https://jjallaire.github.io/hopr/

Wickham, H., M. Cetinka-Rundel and Grolemund, G. 2023. R for data science (2nd edition). Available from https://r4ds.hadley.nz/

Navarro, D., 2019. Learning statistics with R: A tutorial for psychology students and other beginners: version 0.6.1. Adelaide, Australia: University of Adelaide. Available from here

Other books in the library

Campbell, M., 2020. Learn RStudio IDE. *New York: Apress*.

Dinov, I.D., 2018. *Data science and predictive analytics: Biomedical and health applications using R*. Springer.

Mailund, T., 2017. *Beginning Data Science in R: Data Analysis, Visualization, and Modelling for the Data Scientist*. Apress.

Mailund, T., 2019. *R Data Science Quick Reference: A Pocket Guide to APIs, Libraries, and Packages*. Apress.

Matloff, N., 2019. *Probability and statistics for data science: Math+ R+ data*. CRC Press.

Mount, G., 2021. *Advancing into analytics: from Excel to Python and R*, O'Reilly Press.

Please note that there is a wealth of material, much of it freely available on the web or in the library. The above are suggestions but you are encouraged to search for and make use of other sources.

It is important that you learn to become self-reliant and able to access and assimilate material for yourself. Many of the topics necessary for this project will not be covered by lectures. Therefore, you will need to work through the exercises and guidance material provided in labs, via Brightspace or from your own investigations.

ASSESSMENT

This is detailed in the Assessment Blocks CS5801 and CS5802, since technically CS5702 is only a teaching block and the assessments have different module codes. Note that these assessment blocks cover both this module and our 'sister' module Qualitative Data Analysis

EXPECTATIONS OF ARTIFICIAL INTELLIGENCE USE

The University has general guidance on using artificial intelligence in your studies.

Large language models (LLMs) such as ChatGPT, Claude, GitHub Copilot and Gemini all have the potential to generate code, and relevant to this module, R code in response to a text request (prompt). Sometimes the code is of very high standard with helpful accompanying explanations; sometimes the code is not. A difficulty for a learner is to know the difference. Another problem is the behaviour of some LLMs is not static and can vary even with identical prompts.

In this module we will explore how to use LLMs effectively by means of (i) prompt engineering and (ii) careful testing. However, it must be stressed that LLMs are *not* an alternative to learning how to do data science or code.

ADDITIONAL VITAL INFORMATION

Information relating to all aspects of your studies, including procedures of how to inform us of problems you are facing with your studies, how to apply for an extension for your coursework submission, plagiarism and types of other academic misconduct, and other important matters, are available on the University's intranet. The Department assumes that you familiarise yourself with this information, so you will need to look at the relevant intranet pages carefully at various times throughout your studies. The Department also operates within the rules and regulations of the University more generally, and you should also look at what are known as 'Senate Regulations' on the University's web pages. These policies and procedures might change from one academic year to another, and it is in your interest to review them on an on-going basis.